Q1 (2) - Shifting of origin and Rotation of axis

Given, axes are rotated through an angle of 30° To find Position of point $(4,-2\sqrt{3})$ with respect to new axes

We know that, if the position of a point with respect to X and Y -axes is (x, y) and if the axes are rotated through an angle θ, then the new coordinates of the point (x, y) is (x', y') , then

$x'=x\cos\theta+y\sin\theta$

$y'=-x\sin\theta+y\cos\theta$

Here, $\theta=30^{\circ}, x=4,$ and $y=-2\sqrt{3}$

$$\begin{aligned}\therefore x' &= 4\cos 30^{\circ}+(-2\sqrt{3})\sin 30^{\circ}\\ &= \frac{4\sqrt{3}}{2}-2\sqrt{3}\times\frac{1}{2}\\ &= 2\sqrt{3}-\sqrt{3}=\sqrt{3}\\ y' &= -4\sin 30^{\circ}-2\sqrt{3}\cos 30^{\circ}\\ &= -2-3=-5\end{aligned}$$

Hence, the new coordinates of point $(4,-2\sqrt{3})$ is $(\sqrt{3},-5)$.

Q2 (1) - Shifting of origin and Rotation of axis

If point (2,-1) gets reflected about Y -axis, then coordinates of new point will be (-2,-1). Now, the position of point (-2,-1) when axes are rotated through an angle 45° in negative direction (i.e. in clockwise direction).

If X and Y are the new coordinates of the point (-2,-1), then

$$\begin{aligned}X &= x\cos\theta-y\sin\theta\\ Y &= x\sin\theta+y\cos\theta\\ \therefore X &= (-2)\cos 45^{\circ}-(-1)\sin 45^{\circ}\\ &= -2\times\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}=-\frac{1}{\sqrt{2}}\\ Y &= (-2)\sin 45^{\circ}+(-1)\cos 45^{\circ}\\ &= \frac{-2}{\sqrt{2}}-\frac{1}{\sqrt{2}}=-\frac{3}{\sqrt{2}}\end{aligned}$$

Hence, the new coordinates of the point is $\left(-\frac{1}{\sqrt{2}},-\frac{3}{\sqrt{2}}\right)$.

Q3 (2) - Shifting of origin and Rotation of axis

Let origin be shifted at point (h, k) without changing the direction of coordinate axes.

Then, we replace x by $(x+h)$ and y by $(y+k)$ in the equation of given curve, then the transformed equation is

$(x+h)^2+(y+k)^2-4(x+h)+6(y+k)-7=0$

$\Rightarrow \quad x^2+h^2+2hx+y^2+k^2+2ky-4x-4h+6y+6k-7=0$

$x^2+y^2+x(2h-4)+y(2k+6)+h^2+k^2-4h+6k-7=0$

Since, this equation is required to be free from the terms of first degree.

$\therefore 2h-4=0$ and $2k+6=0$

$\Rightarrow h=2$ and $k=-3$

Hence, the point to which the origin be shifted is (2,-3).

Q4 (3) - Angle between Lines, Image and Foot of Perpendicular

If $x-3y=p$ and $ax+2y=q$ are perpendicular, then

$\frac{1}{3}\cdot\left(-\frac{a}{2}\right)=-1$ i.e. $a=6$ ...(i)

If $x-3y=p$ and $ax+y=r$ are perpendicular, then

$\frac{1}{3}\cdot(-a)=-1$ i.e. $\quad a=3$ ...(ii)

$ax+2y=q$ and $ax+y=r$ cannot be perpendicular, as

$-\frac{a}{2}\cdot(-a)=-1 \Rightarrow a^2=-2$ not possible)

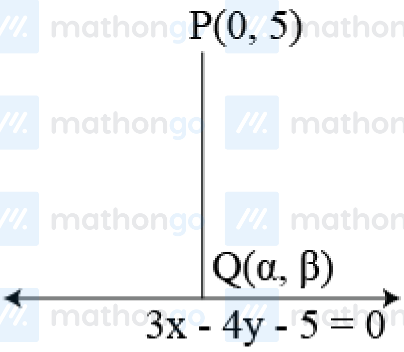
Thus, $a=6$ or $\quad a=3$

$\Rightarrow (a-6)(a-3)=0$

Or $a^2-9a+18=0$

Q5 (4) - Angle between Lines, Image and Foot of Perpendicular

Given equation of line, $3x-4y-5=0$ ...(i)



Let coordinates of foot of perpendicular are $Q(\alpha, \beta)$ ...(ii)

$\therefore \quad 3\alpha - 4\beta = 5$

Slope of line $m_1 = \frac{3}{4}$

Slope of PQ, $m_2 = \frac{\beta - 5}{\alpha}$

Since, PQ is perpendicular on the line $3x - 4y - 5 = 0$

$\therefore m_1 \cdot m_2 = -1$

$\Rightarrow \frac{3}{4}\left(\frac{\beta-5}{\alpha}\right) = -1$

$\Rightarrow \quad 3(\beta - 5) = -4\alpha$

$\Rightarrow \quad 5\alpha + 3\beta = 15 \quad \ldots$(iii)

On solving Eqs. (i) and (iii), we get

$\alpha = 3, \beta = 1$

Hence, coordinates of foot of perpendicular are (3, 1).

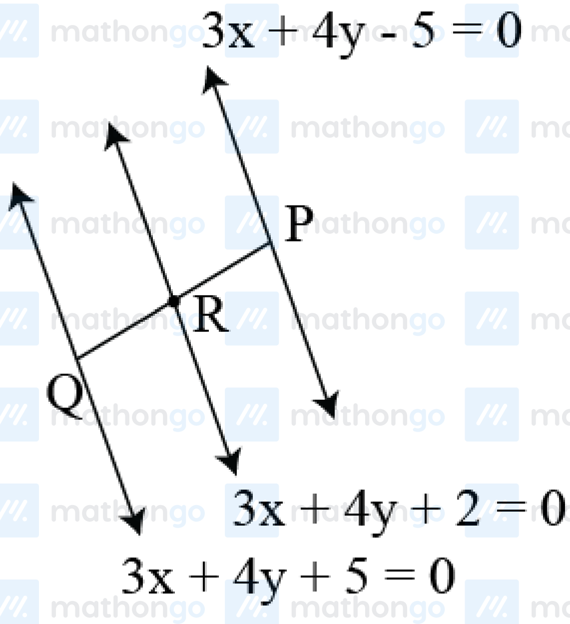
Q6 (2) - Angle between Lines, Image and Foot of Perpendicular

Given, equation of lines, are

$3x + 4y + 2 = 0$ ...(i)

$3x + 4y + 5 = 0$ ...(ii)

$3x + 4y - 5 = 0$ ...(iii)



Any point P on the line can be find by putting $y = 0$ in

$3x + 4y - 5 = 0$

$\Rightarrow 3x + 4 \times 0 - 5 = 0$

$\Rightarrow x = \frac{5}{3}$

$\therefore P\left(\frac{5}{3}, 0\right)$ is a point on the line

$3x + 4y - 5 = 0$

Now, slope of line $3x + 4y + 5 = 0$

$m_1 = -\frac{3}{4}$

Let m_2 be the slope of line PQ. $\because$

$m_1 \cdot m_2 = -1 (\because PQ$ is perpendicular to the given lines)

$\therefore m_2 = \frac{4}{3}$

Equation of PQ, passing through $(5/3, 0)$ having slope $4/3$ is

$$y - 0 = \frac{4}{3}\left(x - \frac{5}{3}\right)$$
$$y = \frac{y}{3}x - \frac{20}{9}$$
$$2x - 3y = 20 \quad \ldots \text{(iv)}$$

On solving equations $3x + 4y + 2 = 0$ and $12x - 9y = 20$

we get coordinates of point R,

i.e. $\left(\frac{62}{75}, \frac{-28}{25}\right)$

On solving equation $3x + 4y + 5 = 0$ and $12x - 9y = 20$ we get the coordinates of point Q,

i.e. $\left(\frac{35}{75}, -\frac{40}{25}\right)$

Let point R divides joint of PQ in the ratio $\lambda : 1$.

$\therefore \frac{62}{75} = \frac{\frac{5}{3} \times \lambda + \frac{35}{75}}{\lambda + 1}$

$$\frac{62}{75} = \frac{125\lambda + 35}{75(\lambda + 1)}$$
$$(\lambda + 1)62 = 125\lambda + 35$$
$$\lambda = \frac{27}{63} = 3 : 7$$

Q7 (4) - Angle between Lines, Image and Foot of Perpendicular

Given, equation of line,

$L : 2x + y = 2$ ..(i)

If the axes are rotated through an angle 45°, then

$$\left.\begin{array}{l} x=X\cos 45^{\circ}-Y\sin 45^{\circ}=\frac{X-Y}{\sqrt{2}} \\ y=X\sin 45^{\circ}+Y\cos 45^{\circ}=\frac{X+Y}{\sqrt{2}} \end{array}\right\} \ldots\text{(ii)}$$

Here, X and Y are coordinates of a point with respect to new axes.

Let p and q be the intercept of line L on the new axes.

$\therefore$ Equation of line L is

$\frac{X}{p}+\frac{Y}{q}=1$ ...(iii)

Substituting the values of x and y in Eq. (i), we get

$2\left(\frac{X-Y}{\sqrt{2}}\right)+\frac{X+Y}{\sqrt{2}}=2$

$\Rightarrow\sqrt{2}(X-Y)+\frac{X+Y}{\sqrt{2}}=2$

$\Rightarrow 2X-2Y+X+Y=2\sqrt{2}$

$\Rightarrow 3X-Y=2\sqrt{2}$

$\Rightarrow\frac{x}{2\sqrt{2}/\sqrt{3}}-\frac{y}{2\sqrt{2}}=1$ ...(iv)

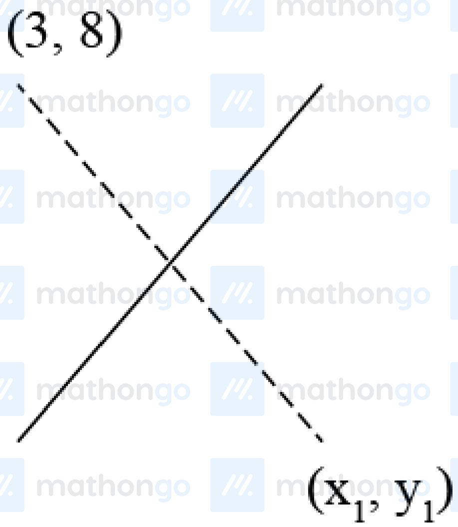
Clearly, Eqs. (iii) and (iv) represent same line.

$\therefore p=\frac{2\sqrt{2}}{3}, q=2\sqrt{2}$

Q8 (2) - Angle between Lines, Image and Foot of Perpendicular

Let reflection of point (3,8) with respect to the line $x+3y=7$ is (x_1, y_1).

$\therefore\ \frac{x_1-3}{1}=\frac{y_1-8}{3}=\frac{-2(3+3\times 8-7)}{10}$



$\Rightarrow\quad x_1-3=\frac{y_1-8}{3}=-4$

$\Rightarrow \quad x_1 - 3 = -4$

$\Rightarrow \quad x_1 = -1$

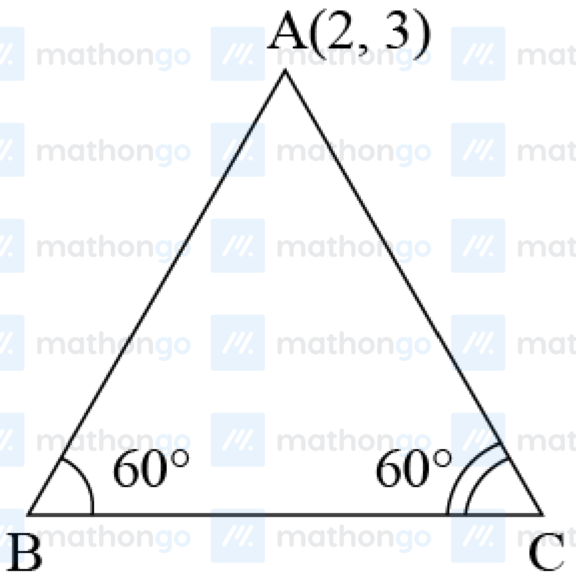
and $\quad y_1 = 8 = -4 \times 3$

$\Rightarrow \quad y_1 = -4$

$\therefore$ Reflection of point (3,8) is (-1,-4)

Q9 (1) - Straight Line Making Given Angle with Line

Let $A(2,3)$ be one vertex and $x+y=2$ be the opposite side of an equilateral triangle. Clearly, remaining two sides pass through the point $A(2,3)$ and make an angle of 60° with $x+y=2$. Let m be the slope of $x+y=2$. Then, $m=-1$. So, the equations of the other two sides are



$y-3=\frac{-1-\tan 60^{\circ}}{1-\tan 60^{\circ}}(x-2)$

and $y-3=\frac{-1+\tan 60^{\circ}}{1+\tan 60^{\circ}}(x-2)$

$\Rightarrow y-3=\frac{-(1+\sqrt{3})}{1-\sqrt{3}}(x-2)$

$=(2+\sqrt{3})(x-2)$

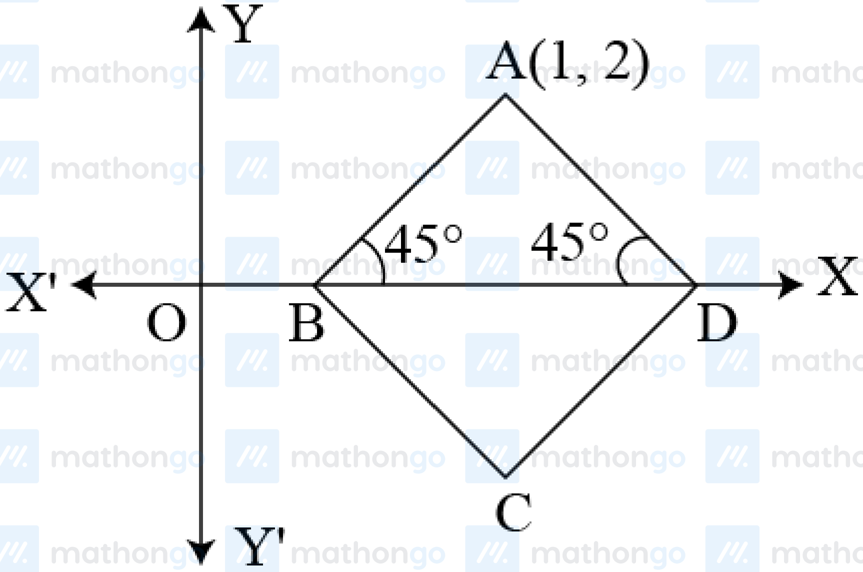
and $y-3=\frac{\sqrt{3}-1}{\sqrt{3}+1}(x-2)$

$=(2-\sqrt{3})(x-2)$

Q10 (3) - Straight Line Making Given Angle with Line

Let $A(1,2)$ be one vertex of square and diagonal BD is along X-axis.

$\therefore \angle ABD = \angle ADB = 45^{\circ}$



Let m be the slope of line BD, then $m = 0$.

$\therefore$ Equation of two sides of square is

$\Rightarrow y - 2 = \frac{0 \pm \tan 45^{\circ}}{1 \mp \tan 45^{\circ} \times 0}(x-1)$

$y - 2 = \pm \tan 45^{\circ}(x-1)$

$y - 2 = \pm(x-1)$

$y - 2 = (x-1)$

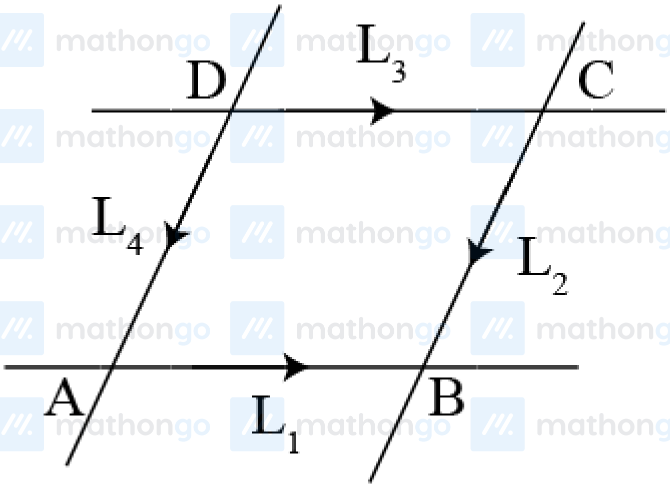
and $y - 2 = -(x-1)$

$\Rightarrow x - y + 1 = 0$

and $x + y - 3 = 0$

Q11 (3) - Straight Line Making Given Angle with Line

Here, equation of diagonals of parallelogram are parallel to



i.e. $L_1 + L_2 = 0$ and $L_1 - L_2 = 0$
$(l+m)x + (m+1)y + 2n = 0$
and $(I-m)x + (m-l)y = 0$

Having slopes -1 and 1.

$\therefore$ Slope of diagonals are $m_{AC} = 1$ and $m_{BD} = -1$

$\Rightarrow \quad m_{AC} \cdot m_{BD} = -1$

Thus, angle between diagonals is 90°.

Q12 (2) - Angle Bisector and Family of lines

Given lines are $\quad 3x - 4y + 7 = 0$ ...(i)

$12x + 5y - 2 = 0$ ...(ii)

Eq. (ii) can be written as

$-12x - 5y + 2 = 0$ ...(iii)

Now. $a_1a_2 + b_1b_2 = 3 \times (-12) + (-4) \times (-5)$
$\Rightarrow = -36 + 20$
$= -16 < 0$

Equation of acute angle bisector

$$\left(\frac{3x-4y+7}{\sqrt{(3)^2+(-4)^2}}\right) = \pm\left(\frac{-12x-5y+2}{\sqrt{(-12)^2+(-5)^2}}\right)$$

[$\because a_1a_2 + b_1b_2 < 0$, origin lies in acute angle, $\therefore$ we take +ve sign]

$\Rightarrow \frac{3x-4y+7}{\sqrt{25}} = \frac{-12x-5y+2}{\sqrt{169}}$

$\Rightarrow \frac{3x-4y+7}{5} = \frac{-12x-5y+2}{13}$

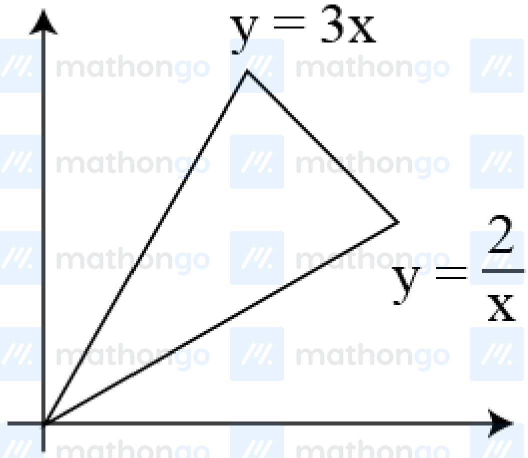
$\Rightarrow 39x - 52y + 91 = -60x - 25y + 10$

$\Rightarrow 99x - 27y + 81 = 0$

$\Rightarrow 11x - 3y + 9 = 0$

Q13 (2) - Angle Bisector and Family of lines

The graph of the equations



$x - 2y = 0$
and $3x - y = 0$

is as shown in the figure given. Since, given point (a, a^2) lies in the shaded region, then $a^2 - \frac{a}{2} > 0$ and $a^2 - 3a < 0$

$\Rightarrow a \in (-\infty, 0) \cup \left(\frac{1}{2}, \infty\right)$ and $a \in (0, 3)$

$\Rightarrow a \in (1/2, 3)$

Q14 (2) - Angle Bisector and Family of lines

Given, $3a + 2b + 4c = 0$

$\frac{3}{4}a + \frac{b}{2} + c = 0$ ...(i)

and family of straight line is

$ax + by + c = 0$ ...(ii)

Subtracting Eq. (i) from Eq. (ii), we get

$a\left(x-\frac{3}{4}\right)+b\left(y-\frac{1}{2}\right)=0$

Which is equation of line passing through the point of intersection of the line

$x-\frac{3}{4}=0$ and $y-\frac{1}{2}=0$

$\therefore x=\frac{3}{4}$ and $y=\frac{1}{2}$

Q15 (2) - Angle Bisector and Family of lines

Given equation, $(2+k)x+(1+k)y=5+7k$

$\Rightarrow \quad (2x+y-5)+k(x+y-7)=0$

which is equation of lines passing through point of intersection of lines

$2x+y-5=0$ ...(i)

and $x+y-7=0$ ...(ii)

On solving Eqs. (i) and (ii), we get

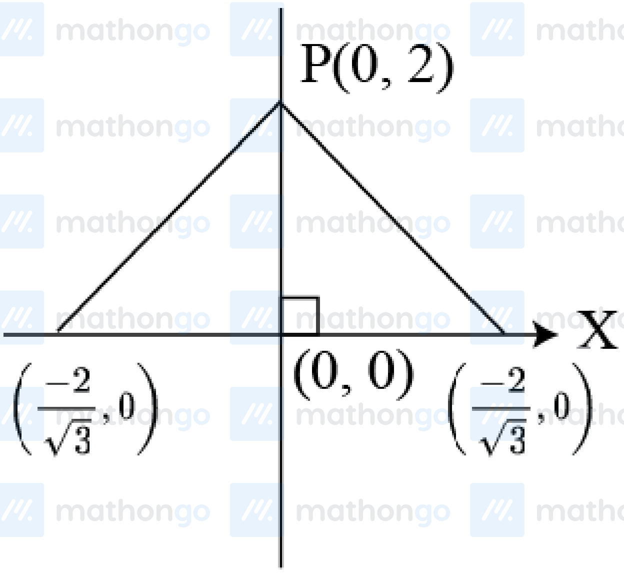
$x=-2, y=9$

Q16 (2) - Angle Bisector and Family of lines

From the following figure,

$y+\sqrt{3}x=2, x>0$

$y-\sqrt{3}x=2, x<0$



$\therefore$ Foot of perpendicular is (0,0)

Q17 (2) - Angle Bisector and Family of lines

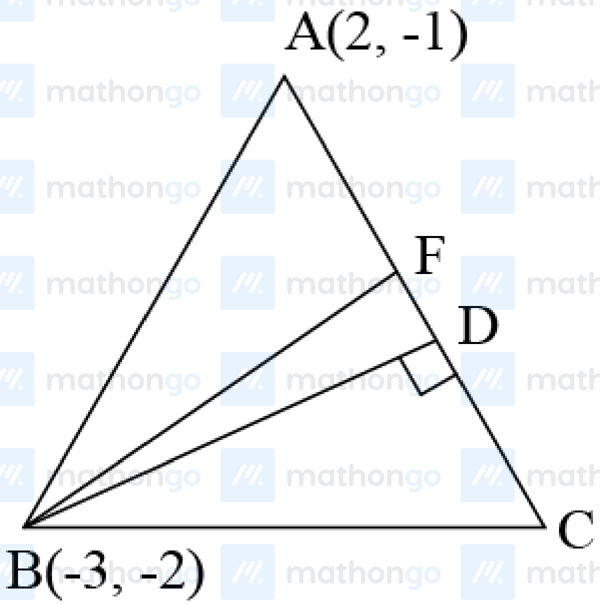
Equation of altitude BD

$7x - 10y + 1 = 0$ ...(i)

Equation of bisector of $\angle B$, i.e. BF

$3x - 2y + 5 = 0$ ...(ii)

On solving Eqs. (i) and (ii), we get The coordinates of point B as $\left(-3, \frac{11}{5}\right)$.



On solving Eqs. (i) and (ii), we get The coordinates of point B as $\left(-3, \frac{11}{5}\right)$.

$$\text{Slope of } AB, m_1 = \frac{-2+1}{-3-2}$$

$$m_1 = \frac{1}{5}$$

Slope of line (ii),

$m = \frac{3}{2}$

Let the slope of line BC is m_2,

$\therefore \quad \frac{m_1 - m}{1 + m_1 m} = -\left(\frac{m_2 - m}{1 + m_2 m}\right)$ ($\because BF$ is equally inclined with AB and BC)

$\Rightarrow \quad \frac{\frac{1}{5} - \frac{3}{2}}{1 + \frac{1}{5} \times \frac{3}{2}} = -\left(\frac{m_2 - \frac{3}{2}}{1 + \frac{3m_2}{2}}\right)$

$\Rightarrow -1 = -\left(\frac{2m_2 - 3}{2 + 3m_2}\right)$

$\Rightarrow \quad 2 + 3m_2 = 2m_2 - 3$

$\Rightarrow m_2 = -5$

Now, equation of BC

$y+2=-5(x+3)$

$\Rightarrow y+2=-5x-15$

$\Rightarrow 5x+y=-17$

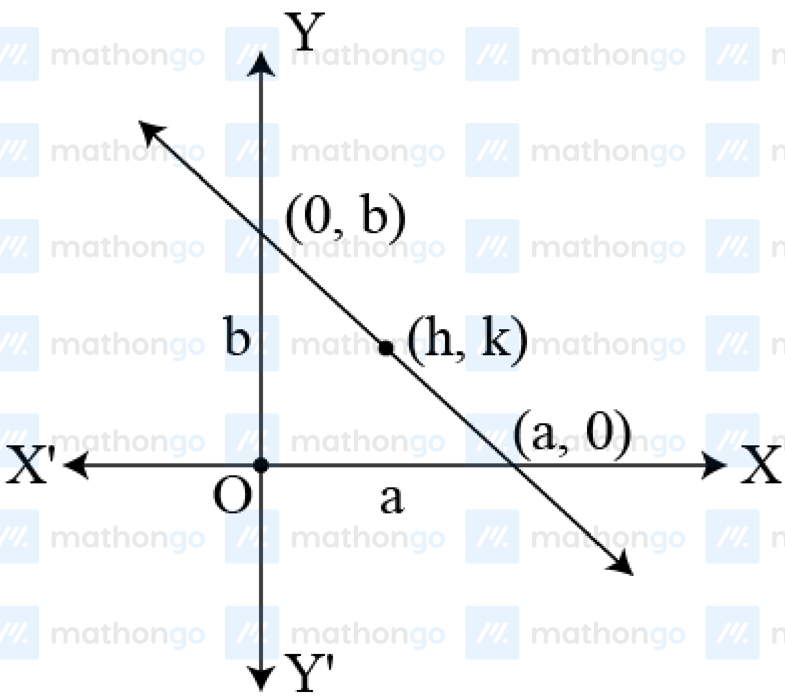
$\Rightarrow 5x+y+17=0$

Q18 (3) - Locus

Given equation of lines are

$\frac{x}{a}+\frac{y}{b}=1$ ...(i)

and $a+b=10$ ...(ii)



Let the locus of mid-point of the intercept between the axes is (h,k)

$\therefore \quad h=\frac{a+0}{2} \Rightarrow a=2h$

and $\quad k=\frac{0+b}{2} \Rightarrow b=2k$

Now. $a+b=10$

$\Rightarrow \quad 2h+2k=10$

$\Rightarrow \quad h+k=5$

or $\quad x+y=5$

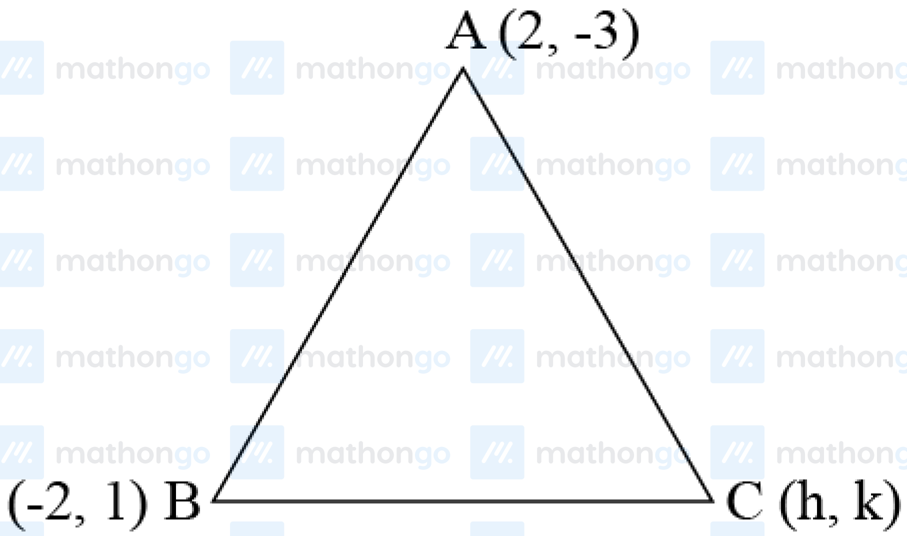
Q19 (3) - Locus

Let locus of C be (h, k)

Now, centroid of ΔABC,

$x = \frac{2-2+h}{3} = \frac{h}{3}$

and $y = \frac{-3+1+k}{3} = \frac{k-2}{3}$, i.e. $\left(\frac{h}{3}, \frac{k-2}{3}\right)$



Since, centroid lies on the line

$2x + 3y = 1$

$\therefore \quad \frac{2h}{3} + \frac{3(k-2)}{3} = 1$

$\Rightarrow \quad \frac{2h}{3} + (k-2) = 1$

$\Rightarrow \quad 2h + 3k - 6 = 3$

$\Rightarrow \quad 2h + 3k = 9$

$\Rightarrow \quad 2x + 3y = 9$

Q20 (4) - Locus

Distance of all the points from (0,0) are 5 units. That means circumcentre of the triangle formed by the given point is (0,0) If $G(h, k)$ we the centroid of triangle, then

$3h = 3 + 5(\cos\theta + \sin\theta)$

and $3k = 4 + 5(\sin\theta - \cos\theta)$

If $H(\alpha, \beta)$ is the orthocentre, then

$OG : GH = 1 : 2$

$\Rightarrow \quad \alpha = 3h, \beta = 3k$

$\Rightarrow \quad \cos\theta + \sin\theta = \frac{\alpha-3}{5}, \sin\theta - \cos\theta = \frac{\beta-4}{5}$

$\Rightarrow \quad \sin\theta = \frac{\alpha+\beta-7}{10}, \cos\theta = \frac{\alpha-\beta+1}{10}$

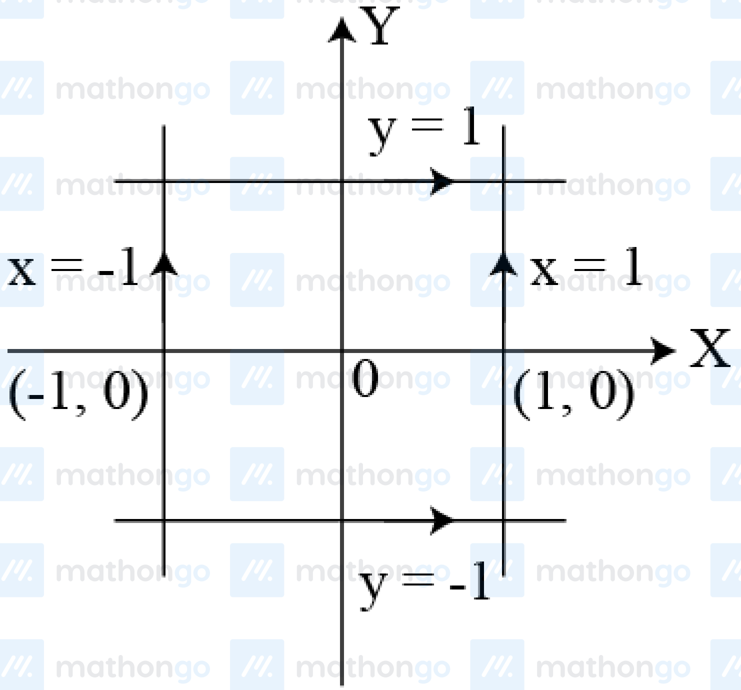
Thus, the locus of (α, β) is

$(x+y-7)^2 + (x-y+1)^2 = 100$

Q21 (2) - Locus

$d(x, y) = \max\{|x|, |y|\}$ (given)

where, $\quad d(x, y) = 1$



$\Rightarrow \quad |x| = 1$ or $|y| = 1$

i.e. a square of area 4 sq units.

Q22 (4) - Pair of straight lines

It is given that

$3x + 4y = 0$

$\Rightarrow y = -\frac{3x}{4}$ is one of the line given by

$6x^2 - xy + 4cy^2 = 0$

$\therefore y=-\frac{3x}{4}$ satisfies it.

Now, $6x^2+\frac{3}{4}x^2+4c\times\frac{9}{16}x^2=-0$

$\Rightarrow\frac{27}{4}+\frac{9c}{4}=0$

$\therefore c=-3$

Q23 (3) - Pair of straight lines

Given, $ax^2+2hxy+by^2=0$

$$m_1+m_2=-\frac{2h}{b} \quad \ldots\text{(i)}$$
$$m_1\cdot m_2=\frac{a}{b} \quad \ldots\text{(ii)}$$
$$m_1:m_2=1:3$$
$$\frac{m_1}{m_2}=\frac{1}{3} \quad \ldots\text{(iii)}$$
$$\Rightarrow 3m_1=m_2 \quad \ldots\text{(iv)}$$

Using Eq.(iv) in Eqs. (ii) and (iii), we get

$4m_1=\frac{-2b}{b}$ and $3m_1^2=\frac{a}{b}$

$\Rightarrow \quad m_1=\frac{-h}{2b}$ and $3m_1^2=\frac{a}{b}$

$\Rightarrow \quad 3\left(\frac{-h}{2b}\right)^2=\frac{a}{b}$

$\Rightarrow 3h^2=4ab$

$\therefore\frac{h^2}{ab}=\frac{4}{3}$

Q24 (2) - Pair of straight lines

Clearly, $y=x$ is one of the lines given by the equation

$ax^2+2hxy+by^2=0$

$\Rightarrow \quad ax^2+2hx^2+bx^2=0 \quad$ (reset $y=x$)

$\therefore \quad a+b=-2h$

Q25 (1) - Pair of straight lines

Given equation, $4x^2+12xy+9y^2=0$

On comparing it with homogeneous equation of second degree, we get

$a=4, b=9, h=6$

Now, $h^2 - ab = (6)^2 - 4 \times 9 = 36 - 36 = 0$

$\therefore$ Lines are real and coincident.

Q26 (1) - Pair of straight lines

Given $a^2x^2 + bcy^2 = a(b+c)xy$
$a^2x^2 + bcy^2 - a(b+c)xy = 0$

On comparing it with homogeneous equation of second degree, we have

$A = a^2, B = bc, H = \frac{-a(b+c)}{2}$
Since, lines are coincident.

$H^2 - AB = \frac{a^2(b+c)^2}{4} - a^2bc = 0$
$\Rightarrow \quad \frac{a^2(b^2+c^2+2bc)}{4} - a^2bc = 0$
$\Rightarrow \quad \frac{a^2(b^2+c^2+2bc)-4a^2bc}{4} = 0$
$\Rightarrow \quad \frac{a^2b^2+a^2c^2+2a^2bc-4a^2bc}{4} = 0$
$\Rightarrow \quad \frac{a^2(b^2+c^2-2bc)}{4} = 0$
$\Rightarrow \quad a^2(b-c)^2 = 0$

Q27 (2) - Pair of straight lines

Given, equation of pair of lines is

$ax^2 - bxy - y^2 = 0$ ...(i)

Given, the lines represented by Eq.(i) makes angle α and β with X -axis.

$\tan\alpha = m_1$ (say)

$\therefore \tan\beta = m_2$ (say)

$$\text{From Eq.(i)}, m_1 + m_2 = \frac{b}{-1} = -b$$
$$m_1 \cdot m_2 = -a$$

Now, $\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha \cdot \tan\beta}$
$= \frac{m_1+m_2}{1-m_1 \cdot m_2} = \frac{-b}{1-(-a)} = \frac{-b}{1+a}$

Q28 (4) - Pair of straight lines

Given, $(p-q)x^2+2(p+q)xy+(q-p)y^2=0$

Given, lines represented by Eq. (i) are perpendicular to each other.

$\therefore$ Coefficient of x^2+ Coefficient of $y^2=0$

$\Rightarrow p-q+q-p=0$

$\therefore p$ and q may have any value.

Q29 (1) - Pair of straight lines

Given, equation of pairs of straight lines are

$ax^2+2hxy+by^2=0 \quad \ldots$(i)

$a'x^2+2h'xy+by^2=0 \quad \ldots$(ii)

Equation of bisector of the angle subtended by the lines represented by Eq. (i) is

$\frac{x^2-y^2}{a-b}=\frac{xy}{h}$

$hx^2-(a-b)xy-hy^2=0$ ...(iii)

Equation of bisector of the angle subtended by the lines represented by Eq. (ii) is

$\frac{x^2-y^2}{a'-b'}=\frac{xy}{h'}$

$h'x^2-(a'-b')\,xy-h'y^2=0$ ...(iv)

Eqs. (iii) and (iv) are same.

$\therefore \frac{h}{h'}=\frac{a-b}{a'-b'}=\frac{h}{h'}$

$\Rightarrow (a-b)h'=(a'-b')\,h$

Q30 (1) - Homogenization of curves

Given, equation of curve

$x^2+y^2=4$ and $y-x=2$

Now, equation of straight line joining point of intersection of above curve and origin is

$x^2+y^2=(y-x)^2$

Q31 (2) - Homogenization of curves

Given, equation of curve is

$(x-k)^2+(y-h)^2=c^2$ ...(i)

and equation of line

$hx+ky=2hk$ ...(ii)

Making the equation of the curve homogeneous with the help of equation of line, we get

$x^2+y^2-2(kx+hy)\left(\frac{hx+ky}{2hk}\right)+\left(h^2+k^2-c^2\right)\left(\frac{hx+ky}{2hk}\right)^2=0$

This is the equation of the pair of lines joining the origin to the point of intersection of the given line and the curves.

These will be at right angle, if coefficient of x^2+ coefficient of $y^2=0$

$$\left(h^2+k^2\right)\left(h^2+k^2-c^2\right)=0$$
$$\Rightarrow h^2+k^2=c^2\left(\because h^2+k^2\neq 0\right)$$

Q32 (3) - Homogenization of curves

Given, equation of circle

$x^2+y^2=a^2$ ...(i)

and equation of line

$$y=mx+c$$
$$\frac{y-mx}{c}=1 \quad \ldots\text{(ii)}$$

Making the equation of the curve homogeneous with the help of that of line, we get

$$x^2+y^2-a^2\left(\frac{y-mx}{c}\right)^2=0$$
$$\Rightarrow x^2+y^2-\frac{a^2}{c^2}\left(y^2+m^2x^2-2mxy\right)=0$$
$$\Rightarrow\left(1-\frac{a^2m^2}{c^2}\right)x^2+\left(1-\frac{a^2}{c^2}\right)y^2+\frac{2a^2m}{c^2}xy=0$$

The lines represented by Eq. (iii) will be perpendicular, if coefficient of x^2+ coefficient of $y^2=0$

$$\Rightarrow\quad 1-\frac{a^2m^2}{c^2}+1-\frac{a^2}{c^2}=0$$
$$\Rightarrow\quad -\frac{a^2}{c^2}\left(m^2+1\right)+2=0$$
$$\Rightarrow\quad -a^2\left(m^2+1\right)=-2c^2$$
$$\therefore\quad a^2\left(m^2+1\right)=2c^2$$